

3. A 230 V series motor is taking 50 A. Resistance of armature and series field windings is 0.2 Ω and 0.1 Ω respectively. Calculate :
 - (a) brush voltage
 - (b) back e.m.f.
 - (c) power wasted in armature
 - (d) mechanical power developed

[(a) 215 V (b) 205 V (c) 500 W (d) 13.74 h.p.] (10.25 kW)
4. Calculate the shaft power of a series motor having the following data; overall efficiency 83.5%, speed 550 r.p.m. when taking 65 A; motor resistance 0.2 Ω , flux per pole 25 mWb, armature winding lap with 1200 conductor.
(15.66 kW)
5. A shunt motor running on no-load takes 5 A at 200 V. The resistance of the field circuit is 150 Ω and of the armature 0.1 Ω . Determine the output and efficiency of motor when the input current is 120 A at 200 V. State any conditions assumed. (89.8%)
6. A d.c. shunt motor with interpoles has the following particulars :
Output power ; 8,952 kW, 440-V, armature resistance 1.1 Ω , brush contact drop 2 V, interpole winding resistance 0.4 Ω shunt resistance 650 Ω , resistance in the shunt regulator 50 Ω . Iron and friction losses on full-load 450 W. Calculate the efficiency when taking the full rated current of 24 A.
(85%)
7. A d.c. series motor on full-load takes 50 A from 230 V d.c. mains. The total resistance of the motor is 0.22 Ω . If the iron and friction losses together amount to 5% of the input, calculate the power delivered by the motor shaft. Total voltage drop due to the brush contact is 2 A. (10.275 kW)
8. A 2-pole d.c. shunt motor operating from a 200 V supply takes a full-load current of 35 A, the no-load current being 2 A. The field resistance is 500 Ω and the armature has a resistance of 0.6 Ω . Calculate the efficiency of the motor on full-load. Take the brush drop as being equal to 1.5 V per brush arm. Neglect temperature rise.
[Rajiv Gandhi Tech. Univ. Bhopal, 2000] (82.63%)

OBJECTIVE TESTS – 29

1. In a d.c. motor, unidirectional torque is produced with the help of
 - (a) brushes
 - (b) commutator
 - (c) end-plates
 - (d) both (a) and (b)
2. The counter e.m.f. of a d.c. motor
 - (a) often exceeds the supply voltage
 - (b) aids the applied voltage
 - (c) helps in energy conversion
 - (d) regulates its armature voltage
3. The normal value of the armature resistance of a d.c. motor is
 - (a) 0.005
 - (b) 0.5
 - (c) 10
 - (d) 100

(Grad. I.E.T.E. June 1987)
4. The E_b/V ratio of a d.c. motor is an indication of its
 - (a) efficiency
 - (b) speed regulation
 - (c) starting torque
 - (d) Running Torque

(Grad. I.E.T.E. June 1987)
5. The mechanical power developed by the armature of a d.c. motor is equal to
 - (a) armature current multiplied by back e.m.f.
 - (b) power input minus losses
 - (c) power output multiplied by efficiency
 - (d) power output plus iron losses
6. The induced e.m.f. in the armature conductors of a d.c. motor is
 - (a) sinusoidal
 - (b) trapezoidal
 - (c) rectangular
 - (d) alternating
7. A d.c. motor can be looked upon as d.c. generator with the power flow
 - (a) reduced
 - (b) reversed
 - (c) increased
 - (d) modified
8. In a d.c. motor, the mechanical output power actually comes from
 - (a) field system
 - (b) air-gap flux
 - (c) back e.m.f.
 - (d) electrical input power
9. The maximum torque of d.c. motors is limited by
 - (a) commutation
 - (b) heating

- (c) speed (d) armature current
10. Which of the following quantity maintains the same direction whether a d.c. machine runs as a generator or as a motor ?
 (a) induced e.m.f. (b) armature current
 (c) field current (d) supply current
11. Under constant load conditions, the speed of a d.c. motor is affected by
 (a) field flux (b) armature current
 (c) back e.m.f. (d) both (b) and (c)
12. It is possible to increase the field flux and, at the same time, increase the speed of a d.c. motor provided its is held constant.
 (a) applied voltage
 (b) torque
 (c) Armature circuit resistance
 (d) armature current
13. The current drawn by a 120 - V d.c. motor of armature resistance 0.5Ω and back e.m.f. 110 V is ampere.
 (a) 20 (b) 240
 (c) 220 (d) 5
14. The shaft torque of a d.c. motor is less than its armature torque because of losses.
 (a) copper (b) mechanical
 (c) iron (d) rotational
15. A d.c. motor develops a torque of 200 N-m at 25 rps. At 20 rps it will develop a torque of N-m.
 (a) 200 (b) 160
 (c) 250 (d) 128
16. Neglecting saturation, if current taken by a series motor is increased from 10 A to 12 A, the percentage increase in its torque is percent
 (a) 20 (b) 44
 (c) 30.5 (d) 16.6
17. If load on a d.c. shunt motor is increased, its speed is decreased due primarily to
 (a) increase in its flux
 (b) decrease in back e.m.f.
 (c) increase in armature current
 (d) increase in brush drop
18. If the load current and flux of a d.c. motor are held constant and voltage applied across its armature is increased by 10 per cent, its speed will
 (a) decrease by about 10 per cent
 (b) remain unchanged
 (c) increase by about 10 per cent
 (d) increase by 20 per cent.
19. If the pole flux of a d.c. motor approaches zero, its speed will
 (a) approach zero
 (b) approach infinity
 (c) no change due to corresponding change in back e.m.f.
 (d) approach a stable value somewhere between zero and infinity.
20. If the field circuit of a loaded shunt motor is suddenly opened
 (a) it would race to almost infinite speed
 (b) it would draw abnormally high armature current
 (c) circuit breaker or fuse will open the circuit before too much damage is done to the motor
 (d) torque developed by the motor would be reduced to zero.
21. Which of the following d.c. motor would be suitable for drives requiring high starting torque but only fairly constant speed such as crushers ?
 (a) shunt (b) series
 (c) compound (d) permanent magnet
22. A d.c. shunt motor is found suitable to drive fans because they require
 (a) small torque at start up
 (b) large torque at high speeds
 (c) practically constant voltage
 (d) both (a) and (b)
23. Which of the following load would be best driven by a d.c. compound motor ?
 (a) reciprocating pump
 (b) centrifugal pump
 (c) electric locomotive
 (d) fan
24. As the load is increased, the speed of a d.c. shunt motor
 (a) increases proportionately
 (b) remains constant
 (c) increases slightly
 (d) reduces slightly
25. Between no-load and full-load, motor develops the *least* torque
 (a) series
 (b) shunt
 (c) cumulative compound
 (d) differential compound

26. The T_a/I_a graph of a d.c. series motor is a
 (a) parabola from no-load to overload
 (b) straight line throughout
 (c) parabola throughout
 (d) parabola upto full-load and a straight line at overloads.
27. As compared to shunt and compound motors, series motor has the highest torque because of its comparatively at the start.
 (a) lower armature resistance
 (b) stronger series field
 (c) fewer series turns
 (d) larger armature current
28. Unlike a shunt motor, it is difficult for a series motor to stall under heavy loading because
 (a) it develops high overload torque
 (b) its flux remains constant
 (c) it slows down considerably
 (d) its back e.m.f. is reduced to almost zero.
29. When load is removed, motor will run at the **highest** speed.
 (a) shunt
 (b) cumulative-compound
 (c) differential compound
 (d) series
30. A series motor is best suited for driving
 (a) lathes
 (b) cranes and hoists
 (c) shears and punches
 (d) machine tools
31. A 220 V shunt motor develops a torque of 54 N-m at armature current of 10 A. The torque produced when the armature current is 20 A, is
 (a) 54 N-m (b) 81 N-m
 (c) 108 N-m (d) None of the above
(Elect. Machines, A.M.I.E. Sec. B, 1993)
32. The d.c. series motor should never be switched on at no load because
 (a) the field current is zero
 (b) The machine does not pick up
 (c) The speed becomes dangerously high
 (d) It will take too long to accelerate.
(Grad. I.E.T.E. June 1988)
33. A shunt d.c. motor works on a.c. mains
 (a) unsatisfactorily (b) satisfactorily
 (c) not at all (d) none of the above
(Elect. Machines, A.M.I.E. Sec. B, 1993)
34. A 200 V, 10 A motor could be rewound for 100 V, 20 A by using as many turns per coil of wire, having the cross-sectional area.
 (a) twice, half
 (b) thrice, one third
 (c) half, twice
 (d) four times, one-fourth

Answers

1. (d) 2. (c) 3. (b) 4. (a) 5. (a) 6. (a) 7. (b) 8. (d) 9. (a) 10. (a) 11. (a)
 12. (d) 13. (a) 14. (d) 15. (a) 16. (b) 17. (b) 18. (c) 19. (b) 20. (c) 21. (c) 22. (d)
 23. (a) 24. (d) 25. (a) 26. (d) 27. (b) 28. (a) 29. (d) 30. (b) 31. (c) 32. (c) 33. (a)
 34. (c).